

Class 1

Tuesday, September 8

Introduction to the Course and Statistical Learning

Upcoming

Reading (complete by Thursday, September 10): [ISL textbook](#) Chapters 1 and 2, and §3.5.

Video (optional) (complete by Thursday, September 10): [1.2 Examples and Framework, Up to 6:30](#); [2.1 Introduction to Regression Models](#); [2.3 Model Selection and Bias–Variance Tradeoff](#)

Homework (due Thursday, September 10 (software installation), September 17 (HW1) at 12:59 p.m.): Install Tophat, R and RStudio. Homework 1 will be posted on Canvas before next class.

Key takeaways

- Goals of statistical learning include prediction, inference, and structure discovery.
- Statistical learning is divided into different paradigms based on the type of data available and data analysis goals. These paradigms include supervised learning and unsupervised learning.
- Supervised learning can be further divided into classification and regression, depending on whether the response variable is quantitative or qualitative.
- Learning involves assumptions. We will often assume that the response is a function of the predictors plus a random error term, and that the training data are i.i.d. copies of the predictor-response pair.

Summary notes

Goals of Statistical Learning. Statistical learning is a broad collection of tools and ideas for learning from data.

Some common examples of ways people want to use data are:

- *Prediction:* Estimate values of an unobserved variable based on observed variables. E.g. the textbook gives an example of predicting wages based on demographic factors.
- *Inference:* Estimate or interpret the relationship between an input and an output. E.g. the textbook gives an example of wanting to understand the relationship between age and wages.
- *Structure discovery:* Cluster similar objects, reduce dimensions, learn a distribution and generate new things. E.g. cluster cells based on gene expression to identify cells that might be the same ‘type’ or share some other biological structure. Sometimes structure discovery is an initial step in a prediction or inference task. Other times it is used to generate new scientific questions or hypotheses.

We will spend the most time on prediction, but will also discuss inference and structure discovery.

Learning Paradigms. Two central learning paradigms are:

- *Supervised learning:* Predicting an output (response) or understanding the relationship between inputs (features, covariates) and outputs using examples of paired inputs and outputs. This is called *regression* if the output is quantitative, and *classification* if it is qualitative. The pairs of inputs and outputs used as examples to build a predictive method are *training data*.

Examples of supervised learning tasks:

- Predict a home’s sale price from square footage, number of bedrooms, and neighborhood.

- Predict a student’s exam score from hours studied.

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- Predict tomorrow’s temperature from today’s weather measurements.

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- Predict which handwritten digit in $\{0, \dots, 9\}$ appears in an image.
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For each example above: (a) write whether it is a classification or regression problem; (b) identify the output variable; and (c) identify the input variables.

- *Unsupervised learning*: Finding structure in unlabeled data, including clustering, dimensionality reduction, and learning a distribution from which new examples can be generated. Examples of unsupervised learning algorithms you might have heard of are *k-means clustering*, *principal component analysis*, and *density estimation*.

Examples of unsupervised learning tasks:

- Group customers into similar segments based on their purchasing behavior.
- Reduce hundreds of gene-expression measurements to a few dimensions that capture major patterns of variation.
- Detect unusual network activity by identifying observations that differ strongly from typical traffic.
- Learn the distribution of handwritten digits and generate new digit images that resemble the training examples.
- There are other types of learning that we won’t cover in this class including *reinforcement learning*, *semi-supervised learning*, *transfer learning*.
- Sometimes the boundary between learning paradigms is blurry. In *self-supervised learning*, an artificial task and artificial labels are constructed from the observed data. For example, a language model can use preceding word (or token) as inputs and the next word (or token) as the response. This looks like a supervised learning problem, but the data was unlabeled.

A Supervised Regression Model and Notation.

- Lowercase letters denote observed values. The value of predictor j for observation i is x_{ij} , the observed predictor vector for observation i is

$$x_i = (x_{i1}, x_{i2}, \dots, x_{ip})^T,$$

and its observed response is y_i .

- The training data are

$$\mathcal{D} = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}.$$

Stacking the predictor vectors gives the $n \times p$ data matrix $\mathbf{X} = (x_{ij})$, and stacking the responses gives $\mathbf{y} = (y_1, \dots, y_n)^T$.

- We will use n for the number of observations and p for the number of predictors. The indices $i = 1, \dots, n$ and $j = 1, \dots, p$ refer to observations and predictors, respectively.
- Uppercase letters denote random quantities. The response is Y , and the predictor vector is

$$X = (X_1, X_2, \dots, X_p).$$

- We assume that the response and predictors satisfy

$$Y = f(X) + \epsilon,$$

where f is a fixed but unknown function and ϵ is a random error term such that

1. X and ϵ are independent, and
 2. $\mathbb{E}[\epsilon] = 0$.
- For the training sample, we make a separate sampling assumption: the n predictor-response pairs are i.i.d. copies of (X, Y) . This is different from assuming that X and ϵ are independent within one observation. This example is made implicitly in various parts of the textbook.